

# Morteza Shoushtari

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## PROFESSIONAL SUMMARY

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Senior Network/System Engineer specializing in AI/HPC network infrastructure at Supermicro, backed by a Ph.D. in Computer and Electrical Engineering. Deep expertise across wire networking (DC and Enterprise), wireless, and cellular network. Passionate about working with great teams and always learning in fast-moving technical environments.

## SKILLS

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### - Networking & Protocols

Wired/Wireless/Data Center Network Architecture & Design | TCP/IP, IPv4/IPv6, FLSM/VLSM | L2 Switching (VLAN, STP, LACP) | L3 Routing (EIGRP, OSPF, BGP, MPLS) | 802.11 (WiFi), 802.15 (Bluetooth, Zigbee) | Cellular Networks (LTE/5G) | Protocol Stack (PHY, MAC, RLC, PDCP, RRC, SDAP)

### - AI/HPC Infrastructure

AI/HPC Network Design & Validation | InfiniBand/Ethernet Networking | RDMA, RoCE | Fiber Optics & Transceivers | High-Speed Interconnects & Cabling Architecture

### - Security & Coding

Cryptographic Algorithms (AES, DES, RSA) | Post-Quantum Cryptosystems | Physical Layer Security | Error-Correction Coding (LDPC, Polar, Turbo, Viterbi)

### - Systems & Tools

Server Operating Systems (Mac, Windows, Linux) | Virtualization Technologies | Network Troubleshooting & Analysis (Wireshark) | Process Automation

### - Programming Languages

C++, MATLAB, Python

### - Certification

Supermicro GPU Service Associate/Specialist/Expert (SGSA, SGSS, SGSE), Supermicro Direct Liquid Cooling Specialist, NVIDIA InfiniBand, CCNA, HCNA, MCITP, Network+, A+

## WORK EXPERIENCE

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### Super Micro Computer, Inc.

CA, USA

*Sr. Network/System Engineer*

*May 2024 – Present*

- Validated network infrastructure for hyperscale AI/HPC deployments, including xAI's Colossus cluster (200k+ NVIDIA H100/H200 GPUs), a 4k-GPU H200 deployment for GBM in Dubai, and a 20k-GPU GB300 deployment for Mistral in Paris, spanning InfiniBand and Ethernet fabrics.
- Reduced deployment errors by 40% by developing and implementing rigorous port-mapping and network-compatibility validation procedures across multi-vendor cabling and transceiver configurations.
- Engineered port connectivity and shuffle solutions (cables/modules) and high-speed interconnect architectures to ensure signal integrity and scalability across racks of varying GPU generations (B200, GB300, VR and AMD MI350/MI450) and interconnect standards.
- Lead network and system validation activities, developing test plans, validation procedures, and technical documentation for network rack components.
- Collaborated with cable, transceiver, and switch vendors to diagnose and resolve complex interoperability issues, minimizing downtime across mission-critical AI infrastructure rollouts.
- Collaborated cross-functionally with datacenter architects, ODM/OEM partners, and deployment teams across the US, Europe and Asia to align on product requirements and ensure successful large-scale rollouts.

### Brigham Young University

Utah, USA

*Graduate Research Assistant*

*Jan 2019 – July 2023*

- Discovered new properties of secrecy coding from the perspectives of information and coding theory.
- Devised a high-speed algorithm to construct the optimal code in Nested Linear Secrecy Codes, enhancing code identification and selection efficiency.

- Conducted a detailed assessment and visualized secrecy problems such as eavesdropping in aeronautical mobile telemetry communication and suggested the use of secrecy coding and specialized version of post-quantum cryptosystems for this type of communication systems.

## HUAWEI

*Network Engineer*

Tehran, Iran

*Aug 2016 – Aug 2017*

- Designed and implemented wired/wireless networks, and performed network maintenance and system upgrades.
- Maximized network performance and increased network availability by 99.99% by implementing a redundant network architecture.
- Performed installations and configurations, and support of various network devices such as switches (Huawei, Cisco), wireless access points (Huawei, Aruba), cameras, and video conferencing systems.

## JYAN

*Network Engineer*

Tehran, Iran

*Jan 2016 – Aug 2016*

- Managed wired communication networks, and performed network maintenance and system upgrades.
- Monitor performance and ensure system availability and reliability.
- Performed installations and configurations of various network devices such as switches (Cisco), wireless access points (Aruba), cameras, and video conferencing systems.

## Persia Cloud

*Network Engineer*

Tehran, Iran

*Jun 2015 – Jan 2016*

- Performed installations and configurations of HP servers, virtual machines (VMs), and Cisco switches.
- Administered SaaS and PaaS layers of the company's cloud platforms (Citrix, IIRAS).
- Managed and configured Microsoft application servers CRM, SharePoint, and Lync.

## EDUCATION

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### Brigham Young University

*Ph.D. in Computer and Electrical Engineering*

UT, USA

*Jan 2019 – July 2023*

### Shiraz University

*M.Sc. in Information Technology*

Fars, Iran

*2011 – 2013*

## SELECTED PUBLICATION

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- “On Best Erasure Wiretap Codes: Equivocation Matrices and Design Principles”, Entropy Journal, 2025.
- “Secrecy Coding for the Binary Symmetric Wiretap Channel via Linear Programming”, IEEE Transactions on Information Forensics and Security, 2025.
- “Optimizing Finite Blocklength Nested Linear Codes: Using the Worst Code to Find the Best Code”, Entropy Journal, 2023.
- Full list available on request

## AWARDS & ACHIEVEMENTS

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- Best paper award, International Telemetry Conference, 2022.
- Second best graduate student paper award, International Telemetry Conference, 2021.
- Third place award, IEEE Intermountain Engineering, Technology, and Computing Conference (IEEE i-ETC), 2023.
- Outstanding IT engineer in Huawei Technologies Company, 2017.

## REFERENCES

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Available upon request.